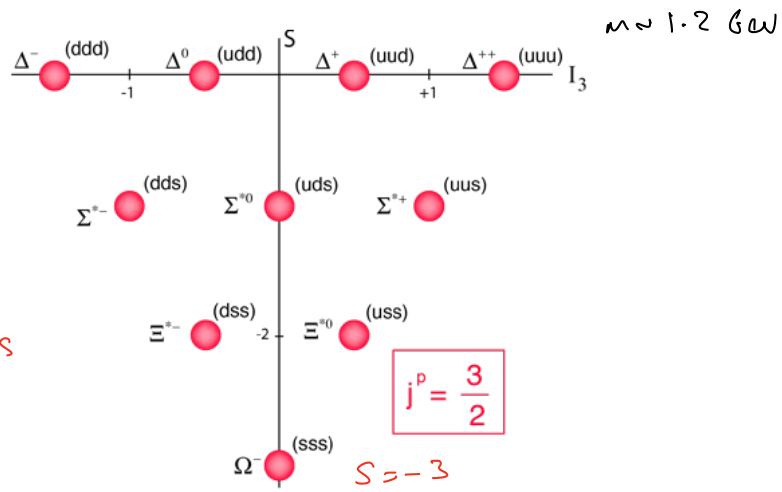
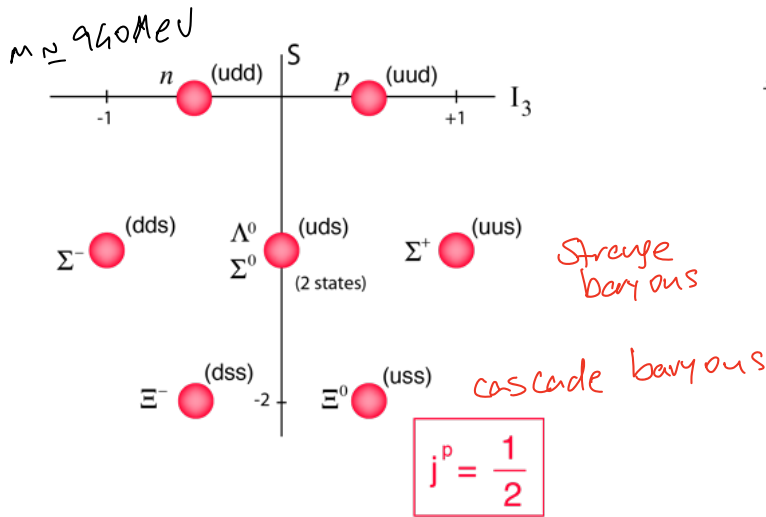


Baryon multiplets



ground state $L=0$.

18 physics states observed.

Spin combinations: $\uparrow\uparrow\uparrow$ $S = 3/2$
 $\uparrow\downarrow\uparrow$ $\uparrow\uparrow\downarrow$ $\downarrow\uparrow\uparrow$ $S = 1/2$

Baryons = $3 \times 3 \times 3 = 1 + 8_{M_{12}} + 8_{M_{23}} + 1_{10} = 27$ states predicted.

Baryons	qqq	J^P	I	I_3	S	$Q^{(1)}$	mass (MeV)
p, n	uud, udd	$\frac{1}{2}^+$	$\frac{1}{2}$	$\frac{1}{2}, -\frac{1}{2}$	0	1, 0	940
Λ	uds	$\frac{1}{2}^+$	0	0	-1	0	1115
$\Sigma^+, \Sigma^0, \Sigma^-$	uus, uds, dds	$\frac{1}{2}^+$	1	1, 0, -1	-1	1, 0, -1	1190
Ξ^0, Ξ^-	uss, dss	$\frac{1}{2}^+$	$\frac{1}{2}$	$\frac{1}{2}, -\frac{1}{2}$	-2	1, 0	1320
$\Delta^{++}, \Delta^+, \Delta^0, \Delta^-$	uuu, uud, udd, ddd	$\frac{3}{2}^+$	$\frac{3}{2}$	$\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$	0	2, 1, 0, -1	1230
$\Sigma^{*+}, \Sigma^{*0}, \Sigma^{*-}$	uus, uds, dds	$\frac{3}{2}^+$	1	1, 0, -1	-1	1, 0, -1	1385
Ξ^{*0}, Ξ^{*-}	uss, dss	$\frac{3}{2}^+$	$\frac{1}{2}$	$\frac{1}{2}, -\frac{1}{2}$	-2	1, 0	1530
Ω^-	sss	$\frac{3}{2}^+$	0	0	-3	-1	1670

μeV
 ~ 100
 $\Delta m = 130$
 $\Delta m \sim 150$

By 1962 all baryons observed except Ω^-

Gell-Mann + Ne'eman predicted existence of Ω^- ($S=-3$)

- predicted mass $\sim 1680 \text{ MeV}$

- long lifetime, weak decay $\tau \sim 10^{-10} \text{ s}$

Decays of Ω^-

Strong decay.

$$\begin{array}{ccc} \Omega^- & \rightarrow & X + Y \\ B & 1 & 1 \quad 0 \\ S & -3 & -3 \quad 0 \\ S & -3 & -2 \quad -1 \end{array}$$

$\Rightarrow X$ baryon
 Y meson

NO OTHER $S=-3$ particle

$$X = \Xi, \Xi^* (S=-2)$$

Y : meson with $S=-1$

$$\begin{array}{cc} S\bar{d} & S\bar{u} \\ \bar{K}^0 & K^- \end{array}$$

pseudo scalar mesons $S=0$

in principle also K^{*0}, K^{*-}
check J is conserved.

$$\Omega^- \rightarrow \Xi^{(*)-} K^0$$

$$Q \quad -1 \quad -1 \quad 0$$

$$\Omega^- \rightarrow \Xi^{(*)0} K^{(*)-}$$

$$J \quad \frac{3}{2} \quad (\frac{3}{2}, \frac{1}{2})$$

$$\Omega^- \rightarrow \Xi^0 K^-, \Xi^{*0} K^-, \Xi^0 K^{*-}, \Xi^{*0} K^{*-}$$

$$J \quad \frac{3}{2} \quad \frac{1}{2} \quad 0 \quad \frac{3}{2} \quad 0 \quad \frac{1}{2} \quad 1 \quad \frac{3}{2} \quad 1$$

Decay possible if $Q \geq 0$

$$Q = m_{\Omega} - m_{\Xi^{(*)}} - m_{K^{(*)}}$$

$$\begin{array}{l} 1680 - 1320 - 500 \text{ MeV } K \\ 1530 - 900 \text{ MeV } K^* \end{array} < 0$$

$\Rightarrow$ strong interaction kinematically not allowed

EM decay

$$\Omega^- \rightarrow X + Y$$

$$Q \quad -1 \quad -1 \quad 0$$

$$B \quad 1 \quad 1 \quad 0$$

$$S \quad -3 \quad -3 \quad 0$$

$$\Omega^- \rightarrow X + \gamma$$

X : baryon with $Q=-1$

no other X with $S=-3$

$\Rightarrow$ EM not possible

$\Rightarrow$ only weak decay allowed $\Rightarrow \tau \sim 10^{-10} \text{ s}$

Weak decays

look at $\Delta S = 1$ decays.

$$Q = m_\Sigma - m_Y - m_X \geq 0$$

$$\begin{array}{ccc} \Sigma^- & \rightarrow & X + Y \\ B & 1 & 1 \quad 0 \\ Q & \begin{bmatrix} -1 & -1 & 0 \\ & 0 & -1 \end{bmatrix} \end{array}$$

$\Rightarrow$ X: baryon
Y: meson.

$$S \begin{bmatrix} -3 & -2 & 0 \\ -3 & 0 & -2 \\ -3 & -1 & -1 \end{bmatrix}$$

No meson with $S = -2$

$$m_{\Xi^*} > m_{\Xi} \Rightarrow \Xi \text{ preferred because } Q_{\Xi} > Q_{\Xi^*}$$

$$Q = m_\Sigma - m_{\Xi} - m_Y = 1680 - 1320 - m_Y = 360 \text{ MeV} - m_Y \geq 0$$

$$Y: \text{meson, } S=0, m < 360 \text{ MeV} \Rightarrow Y = \pi^{-10}$$

possible decays:

$$\Sigma^- \rightarrow \Xi^0 \pi^-$$

$$\Xi^- \pi^0$$

the option $S = -3 \rightarrow -1 \quad -1$

$$\Sigma^- \rightarrow X^0/X^- + K^-/K^0$$

1680

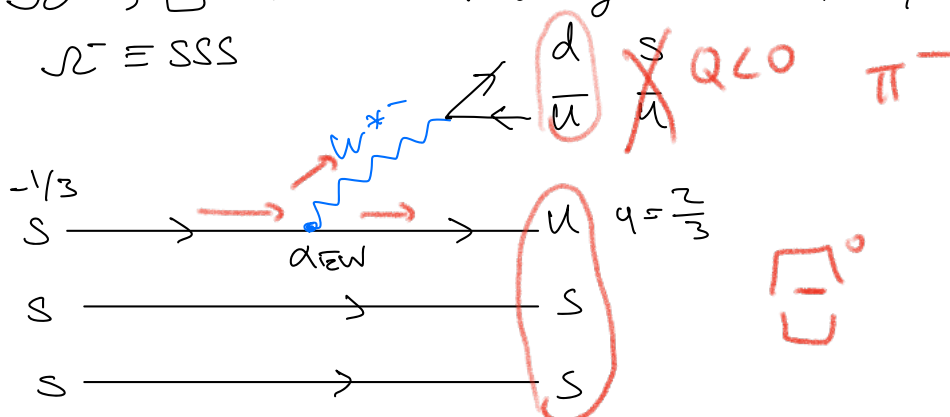
500 MeV.

$$X: S = -1, B = 1, m < 1680 - 500 = 1180$$

$$\Sigma^- \rightarrow \Lambda^0 + K^- \quad Q \text{ small} = 65 \text{ MeV}$$

$$\Sigma^- \rightarrow \Xi^0 \pi^- \quad \text{weak decay mediated by } W^\pm$$

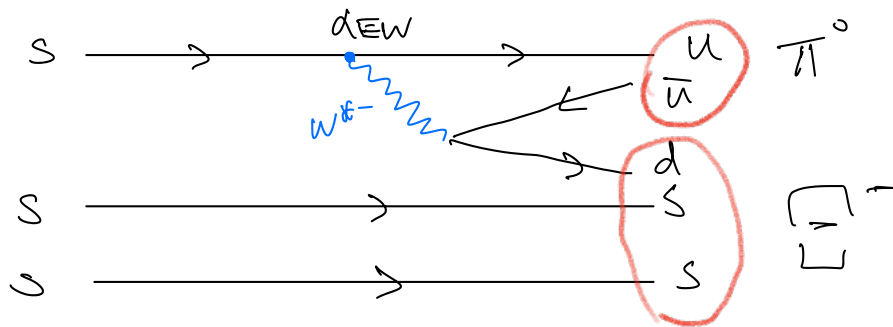
$$\Sigma^- \equiv SSS$$



Spectator diagram

$$BF(\Sigma^- \rightarrow \Xi^0 \pi^-) = 24\%$$

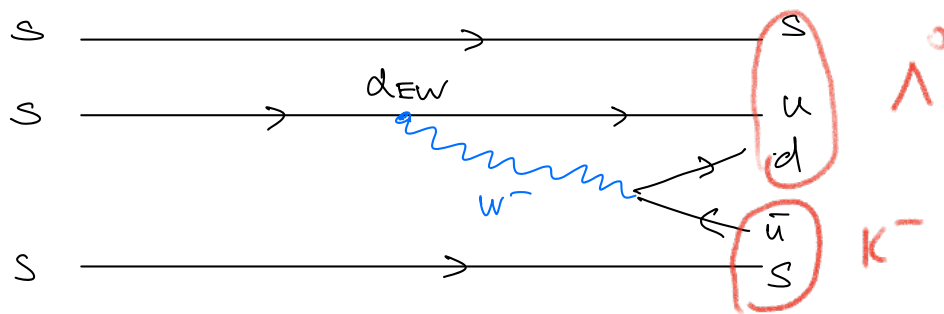
$$\Sigma^- \rightarrow \Xi^- \pi^0$$



$$BF(\Sigma^- \rightarrow \Xi^- \pi^0) = 9\%$$

Internal diagram

$$\Sigma^- \rightarrow \Lambda^0 K^-$$



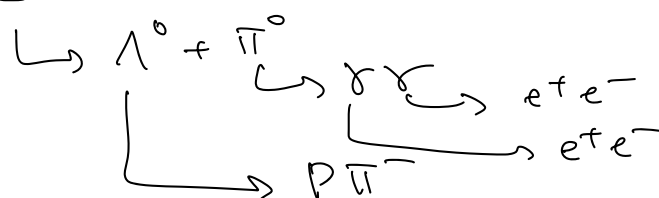
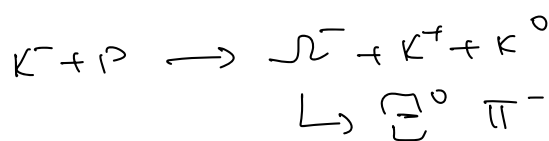
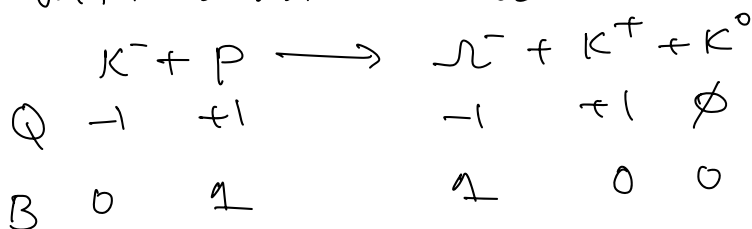
$$BF(\Sigma^- \rightarrow \Lambda^0 K^-) = 68\%$$

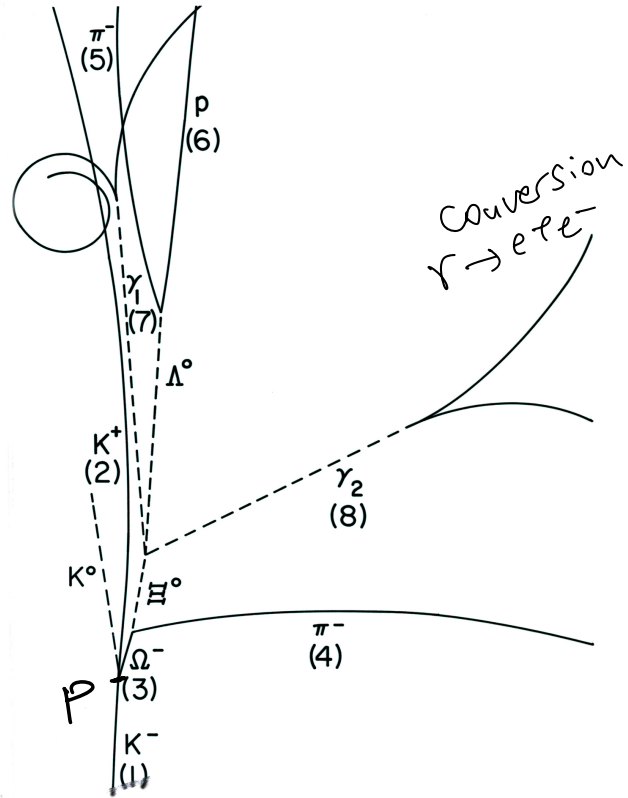
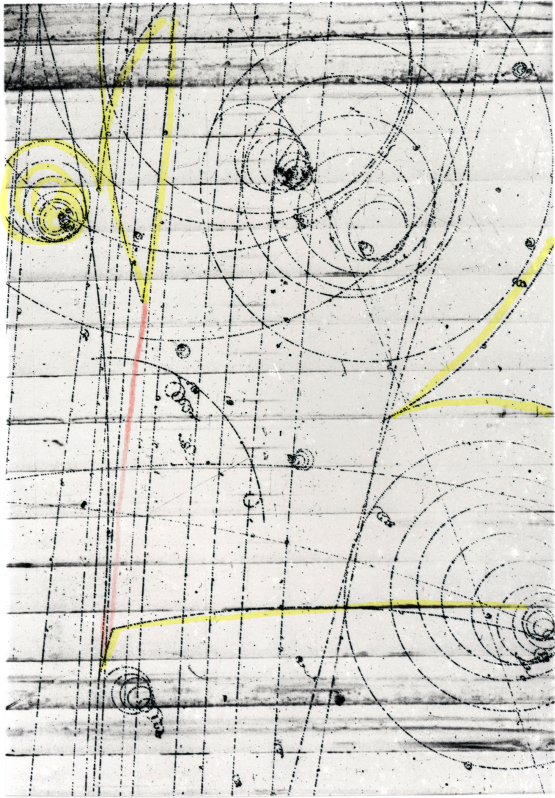
$$\Gamma \propto |M|^2 \rho(E)$$

Ω^- DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
$\Gamma_1 \quad \Lambda K^-$	$(67.7 \pm 0.7) \%$	S=1.5
$\Gamma_2 \quad \Xi^0 \pi^-$	$(24.3 \pm 0.7) \%$	
$\Gamma_3 \quad \Xi^- \pi^0$	$(8.55 \pm 0.33) \%$	
$\Gamma_4 \quad \Xi^- \pi^+ \pi^-$	$(3.7^{+0.7}_{-0.6}) \times 10^{-4}$	CL=90%
$\Gamma_5 \quad \Xi(1530)^0 \pi^-$	$< 7 \times 10^{-5}$	
$\Gamma_6 \quad \Xi^0 e^- \bar{\nu}_e$	$(5.6 \pm 2.8) \times 10^{-3}$	
$\Gamma_7 \quad \Xi^- \gamma$	$< 4.6 \times 10^{-4}$	CL=90%
$\Delta S = 2$ forbidden (S_2) modes		
$\Gamma_8 \quad \Lambda \pi^-$	$S_2 \quad < 2.9 \times 10^{-6}$	CL=90%

Experimental proof @ Brookhaven National Lab.
with bubble Chambers.





Evidence for $\Omega^- \Rightarrow SU(3)_F$ works.

$$\Delta^{++} \equiv uuu$$

$$S = 3/2$$

$$\Delta^{++} = u \uparrow u \uparrow u \uparrow$$

$$\Omega^- = s \downarrow s \downarrow s \downarrow$$

$$\Delta^- = d \uparrow d \uparrow d \uparrow$$

$S = 3/2 \Rightarrow$ baryons are fermions

$$\psi_{\Delta^{++}} = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{flavor}} \times \psi_{\text{color}}$$

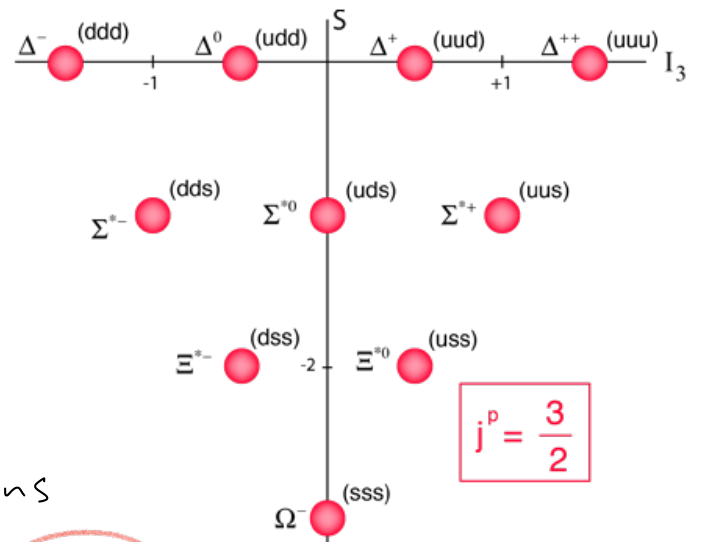
$$L=0 \quad (-1)^L \quad S=3/2$$

under exchange
 $1 \leftrightarrow 2$
 $2 \leftrightarrow 3$
 $1 \leftrightarrow 3$

under exchange
 $q_1 \leftrightarrow q_2$
 $q_1 \leftrightarrow q_3$
 $q_2 \leftrightarrow q_3$

(A)

New quantum number color



How many colors?

$$\Delta^{++} = \uparrow u_R \uparrow u_R \uparrow u_R$$

useless.

1 color only

2 colors.

$$\uparrow u_R \uparrow u_B \uparrow u_R$$

R, B colors.

symm. under $1 \leftrightarrow 3$ exchange

3 colors: RGB

$$\Delta^{++} = \uparrow u_R \uparrow u_G \uparrow u_B$$

$$\psi_{\text{color}} = [RGB + BRG + GBR + \dots - RBG - GRB - BGR] \frac{1}{\sqrt{6}}$$

Completely
anti-symm.
color state.

$\Rightarrow$ $SU(3)$ color symmetry

$$\text{Generators} = 9 - 1 = 8$$

2 diagonal generators.

$$SU(3)_F$$

$$Q = I_3 + \frac{B+S+...}{2} = I_3 + \frac{Y}{2}$$

Y : hypercharge.

2 diagonal generators.

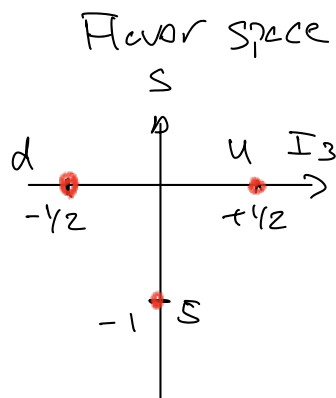
I_3
Strangeness/

$$Y = B + S = \frac{1}{3} + S.$$

$$Y_u = \frac{1}{3} + \frac{1}{2} = \frac{3+2}{6} = \frac{5}{6}$$

$$Y_d = -\frac{1}{3} + \frac{1}{2} = \frac{-2+3}{6} = \frac{1}{6}$$

$$Y_s = \frac{1}{3} - 1 = -\frac{2}{3}$$

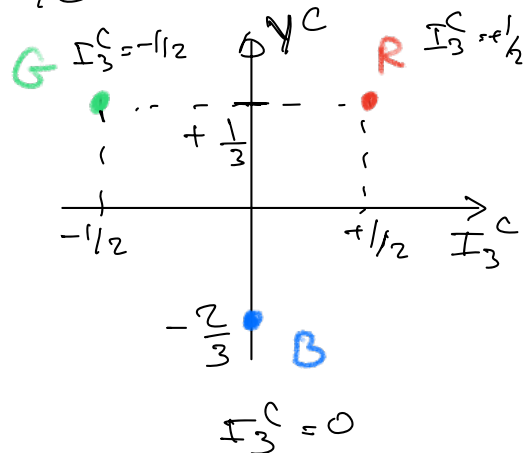


$SU(3)$ color

2 diagonal generators

I_3 : color isospin

Y_C : color hypercharge.



$$R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, G = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

quarks have 3 colors R, G, B

anti-quarks have 3 anti-colors $\bar{R}, \bar{G}, \bar{B}$

Quantum Chromodynamics (QCD)

interaction between quarks by exchange of gluons.
having color charge.

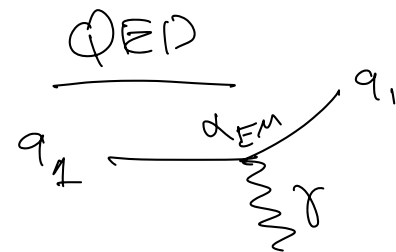
gluons: mediators of QCD Gauge theory
8 generators $\Rightarrow$ 8 gluons.

color: new kind of charge for quarks

Hypothesis/conjecture:

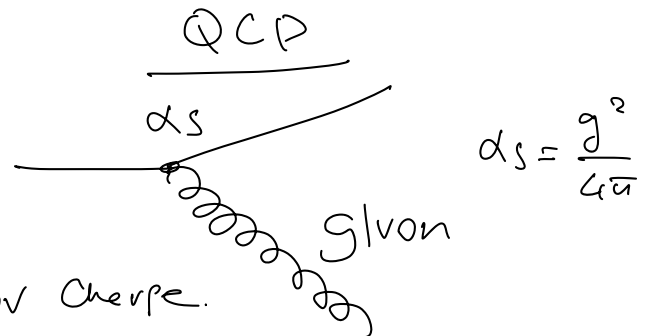
physical particles in nature are colorless.

net/total color charge $= \emptyset$



$$q = -e \quad e^-$$
$$q = +e \quad e^+$$
$$Q_\gamma = 0$$

$$\alpha = \frac{e^2}{4\pi}$$



$$\alpha_s = \frac{g^2}{4\pi}$$

g : color charge.

8 gluons.

massless, $S=1$ (vector bosons)

gluons carry color charge.